

Somaiya Vidyavihar University
K J Somaiya School of Engineering

Batch: **Roll No.:**
Experiment / assignment / tutorial No. 02
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

Title: Amplitude Modulation

Objectives:

1. To generate Amplitude modulated waveform (virtual lab and hardware)
2. To measure the modulation index

OUTCOME:

1. analyze and compare analog modulation schemes.
2. understand noise effect on the performance of communication systems.

Part-1: Amplitude Modulation using Virtual Lab

- Visit the link: <https://kcgcollege.ac.in/Virtual-Lab/Electronics-and-Communication-Engineering/index.html>
- Follow step by step procedure of the experiment and perform
- Output: Take screenshot of input and output waveforms
- Write conclusion based on your learning

Part-2: Amplitude Modulation using Hardware

• **Procedure:**

1. Make the connections as per given circuit diagram.
2. Connect the modulating signal and carrier signal.
3. Observe the AM output for modulation index, $m=1$, $m<1$ and $m>1$ and calculate modulation index by, $m = (V_{\max} - V_{\min}) / (V_{\max} + V_{\min})$.
4. Observe the waveforms for $m=1$, $m>1$ and $m<1$.
5. Give modulated wave to the Spectrum Analyzer to observe AM spectrum.
6. Plot the AM waveforms for $m=1$, $m>1$ and $m<1$ on the graph paper.

• **Circuit diagram: Draw circuit diagram**

• **Input:**

Sr. No.	Signal	Amplitude (Peak)	Frequency
1.	Modulating Signal		
2.	Carrier Signal		

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- **Observation Table (Time Domain):**

Cases	V_C (p-p)	V_m (p-p)	V_{max}	V_{min}	m	% m
$m < 1$						
$m = 1$						
$m > 1$						

- **Calculations:** Show calculations for $m < 1$, $m = 1$, $m > 1$
- **Waveforms:** Sketch neat and labelled modulating signal, carrier signal and amplitude modulated wave for $m < 1$, $m = 1$ and $m > 1$ with proper scale.

- **Post lab activity:**

1. Why must the modulation index be kept less than or equal to 1 in amplitude modulation? What physical change did you observe in the waveform when this condition was violated?
2. Why is the carrier frequency chosen much higher than the modulating signal frequency in the AM experiment? What would happen if both frequencies were comparable?
3. What is the effect of increasing the modulating signal amplitude while keeping the carrier amplitude constant? How does this reflect in the modulation index?
4. Why does AM transmission consume more power compared to the information actually transmitted? Identify which component of the AM signal is responsible.
5. In the frequency spectrum observed (or inferred), why do two sidebands appear and why are they symmetrically placed around the carrier?

Signature of faculty in-charge